Honeywell Solstice® N40 Refrigerant (R-448A)

000000017419

Version 2.4	Date of first issue 09/04/2013	Revision Date 02/18/2026	Print Date 06/23/2026

1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product information

Trade name	:	Honeywell Solstice® N40 Refrigerant (R-448A)
Number	:	000000017419
Recommended use of the chemical and restrictions on use	:	Refrigerant
Manufacturer or supplier's details	:	Solstice Trading (Shanghai) Co., Ltd. No.69 Xi Ya Rd, Shanghai, 31 200131 China
For further information, please contact:	:	800-522-8001 +1-973-455-6300 (Monday-Friday, 9:00am-5:00pm)
In case of emergency call	:	**Medical: 1 866-479-8788 or +1 303-739-1378**
	:	**Transportation (CHEMTREC): 1-800-424-9300 or +1-703-527-3887**
		In Thailand: 1800014808

2. HAZARDS IDENTIFICATION

Classification of the substance or mixture

Classification of the substance or mixture : Gases under pressure, Liquefied gas

GHS Label elements, including precautionary statements

Symbol(s) :



Signal word	:	Warning
Hazard statements	:	Contains gas under pressure; may explode if heated.
Precautionary statements	:	**Prevention:** Use personal protective equipment as required. **Storage:** Protect from sunlight. Store in a well-ventilated place.
Other hazards which do not result in classification	:	Causes asphyxiation in high concentrations. The victim will not realize that he/she is suffocating. May cause cardiac arrhythmia. May cause frostbite. May irritate skin. May irritate eyes.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical nature : Mixture

Chemical name	CAS-No.	Concentration
Difluoromethane Difluoromethane	75-10-5	26.00 %
Pentafluoroethane Pentafluoroethane	354-33-6	26.00 %
1,1,1,2-Tetrafluoroethane 1,1,1,2-Tetrafluoroethane	811-97-2	21.00 %
2,3,3,3-Tetrafluoroprop-1-ene 2,3,3,3-Tetrafluoroprop-1-ene	754-12-1	20.00 %
trans-1,3,3,3-Tetrafluoroprop-1-ene trans-1,3,3,3-Tetrafluoroprop-1-ene	29118-24-9	7.00 %

4. FIRST AID MEASURES

General advice	:	First aider needs to protect himself. Move out of dangerous area. Take off all contaminated clothing immediately.
Inhalation	:	Move to fresh air. If breathing is irregular or stopped, administer artificial respiration. Use oxygen as required, provided a qualified operator is present. Call a physician. Do not give drugs from adrenaline-ephedrine group.
Skin contact	:	After contact with skin, wash immediately with plenty of water. If there is evidence of frostbite, bathe (do not rub) with lukewarm (not hot) water. If water is not available, cover with a clean, soft cloth or similar covering. If symptoms persist, call a physician.
Eye contact	:	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In case of frostbite water should be lukewarm, not hot. If symptoms persist, call a physician.
Ingestion	:	Unlikely route of exposure. As this product is a gas, refer to the inhalation section. Do not induce vomiting without medical advice. Call a physician immediately.
Notes to physician	:	Because of the possible disturbances of cardiac rhythm, catecholamine drugs, such as epinephrine, should be used with special caution and only in situations of emergency life support. Treatment of overexposure should be directed at the control of symptoms and the clinical conditions. Treat frost-bitten areas as needed.

5. FIREFIGHTING MEASURES

Suitable extinguishing media	:	The product is not flammable. Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Specific hazards during firefighting	:	Contents under pressure. This product is not flammable at ambient temperatures and atmospheric pressure. However, this material can ignite when mixed with air under pressure and exposed to strong ignition sources. Container may rupture on heating. Cool closed containers exposed to fire with water spray. Do not allow run-off from fire fighting to enter drains or water courses. Vapours are heavier than air and can cause suffocation by reducing oxygen available for breathing. Fire may cause evolution of: Halogenated compounds Hydrogen fluoride Carbon oxides Carbonyl halides
Special protective equipment for firefighters	:	In the event of fire and/or explosion do not breathe fumes. Wear self-contained breathing apparatus and protective suit. No unprotected exposed skin areas.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	:	Immediately evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Wear personal protective equipment. Unprotected persons must be kept away. Remove all sources of ignition. Avoid skin contact with leaking liquid (danger of frostbite).

Ventilate the area.
After release, disperses into the air.
Vapours are heavier than air and can cause suffocation by reducing oxygen available for breathing.
Avoid accumulation of vapours in low areas.
Unprotected personnel should not return until air has been tested and determined safe.
Ensure that the oxygen content is >= 19.5%.

Environmental precautions : Prevent further leakage or spillage if safe to do so.
The product evaporates readily.

Methods and materials for containment and cleaning up : Ventilate the area.

7. HANDLING AND STORAGE

Handling

Precautions for safe handling : Handle with care.
Avoid inhalation of vapour or mist.
Do not get in eyes, on skin, or on clothing.
Wear personal protective equipment.
Use only in well-ventilated areas.
Pressurized container. Protect from sunlight and do not expose to temperatures exceeding 50 °C.
Follow all standard safety precautions for handling and use of compressed gas cylinders.
Use authorized cylinders only.
Protect cylinders from physical damage.
Do not puncture or drop cylinders, or expose them to open flame or excessive heat.
Do not pierce or burn, even after use. Do not spray on a naked flame or any incandescent material.
Do not remove screw cap until immediately ready for use.
Always replace cap after use.

Advice on protection against fire and explosion : The product is not flammable.
Can form a combustible mixture with air at pressures above atmospheric pressure.

Storage

Conditions for safe storage, including any incompatibilities	:	Pressurized container: protect from sunlight and do not expose to temperatures exceeding 50 °C. Do not pierce or burn, even after use. Keep containers tightly closed in a dry, cool and well-ventilated place. Storage rooms must be properly ventilated. Ensure adequate ventilation, especially in confined areas. Protect cylinders from physical damage.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Value	Control parameters	Update	Basis
Difluoromethane Difluoromethane	75-10-5	TWA : Time weighted average	(1,000 ppm)	1994	Solstice:Limit established by Solstice Advanced Materials
		TWA : Time weighted average	2,200 mg/m3 (1,000 ppm)	2007	WEEL:US Workplace Environmental Exposure Level
Pentafluoroethane Pentafluoroethane	354-33-6	TWA : Time weighted average	(1,000 ppm)		Solstice:Limit established by Solstice Advanced Materials

		TWA : Time weighted average	4,900 mg/m3 (1,000 ppm)	2007	WEEL:US Workplace Environmental Exposure Level
1,1,1,2-Tetrafluoroethane 1,1,1,2-Tetrafluoroethane	811-97-2	TWA : Time weighted average	(1,000 ppm)		Solstice:Limit established by Solstice Advanced Materials
		TWA : Time weighted average	4,240 mg/m3 (1,000 ppm)	2007	WEEL:US Workplace Environmental Exposure Level
2,3,3,3-Tetrafluoroprop-1-ene 2,3,3,3-Tetrafluoroprop-1-ene	754-12-1	TWA : Time weighted average	(500 ppm)	03 15 2010	Solstice:Limit established by Solstice Advanced Materials
		STEL : Short term exposure limit	(1,500 ppm)	03 15 2010	Solstice:Limit established by Solstice Advanced Materials

		TWA : Time weighted average	(500 ppm)	2009	WEEL:US Workplace Environmental Exposure Level

trans-1,3,3,3-Tetrafluoroprop-1-ene trans-1,3,3,3-Tetrafluoroprop-1-ene	29118-24-9	TWA : Time weighted average	(800 ppm)	31.03.11	Solstice:Limit established by Solstice Advanced Materials

		TWA : Time weighted average	(800 ppm)	2022	WEEL:US Workplace Environmental Exposure Level

Appropriate engineering controls

General room ventilation is adequate for storage and handling.
Perform filling operations only at stations with exhaust ventilation facilities.

Individual protection measures, such as personal protective equipment (PPE)

Respiratory protection : In case of insufficient ventilation, wear suitable respiratory equipment.
Wear a positive-pressure supplied-air respirator.
Vapours are heavier than air and can cause suffocation by reducing oxygen available for breathing.
For rescue and maintenance work in storage tanks use self-contained breathing apparatus.

Hand protection : Leather gloves
In case of contact through splashing:
Protective gloves
Neoprene gloves

Polyvinyl alcohol or nitrile- butyl-rubber gloves

Eye protection	:	Wear as appropriate: Safety glasses with side-shields If splashes are likely to occur, wear: Goggles or face shield, giving complete protection to eyes
Skin and body protection	:	Avoid skin contact with leaking liquid (danger of frostbite). Wear cold insulating gloves/ face shield/ eye protection.
Hygiene measures	:	Handle in accordance with good industrial hygiene and safety practice. Ensure adequate ventilation, especially in confined areas. Avoid contact with skin, eyes and clothing. Remove and wash contaminated clothing before re-use. Keep working clothes separately.
Protective measures	:	Do not breathe vapor. Avoid contact with skin, eyes and clothing. Ensure that eyewash stations and safety showers are close to the workstation location.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	:	Liquefied gas
Colour	:	colourless
Odour	:	slight ether-like
Odour Threshold	:	Note: No data available
pH	:	Note: neutral
Melting point/ range	:	Note: No data available
Boiling point/boiling range	:	Note: No data available

Flash point	:	Note: Not applicable
Evaporation rate	:	Note: No data available
Vapour pressure	:	1,398 kPa at 21.1 °C(70.0 °F)
Vapour density	:	Note: No data available
Density	:	Note: No data available
Water solubility	:	Note: No data available
Partition coefficient: n-octanol/water	:	Note: No data available
Auto-ignition temperature	:	628 °C
Viscosity, dynamic	:	Note: No data available
Viscosity, kinematic	:	Note: No data available
Oxidizing properties	:	The substance or mixture is not classified as oxidizing.

10. STABILITY AND REACTIVITY

Chemical stability	:	Stable under normal conditions.
Possibility of hazardous reactions	:	Hazardous polymerisation does not occur.

Conditions to avoid	:	Pressurized container. Protect from sunlight and do not expose to temperatures exceeding 50 °C. Decomposes under high temperature. Some risk may be expected of corrosive and toxic decomposition products. Can form a combustible mixture with air at pressures above atmospheric pressure. Do not mix with oxygen or air above atmospheric pressure.
Incompatible materials to avoid	:	Potassium Calcium Powdered metals Finely divided aluminium Finely divided magnesium Zinc
Hazardous decomposition products	:	Halogenated compounds Hydrogen fluoride Carbonyl halides Carbon oxides

11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure	:	Inhalation, Eye contact, Skin contact
Acute oral toxicity	:	Note: No data available
Acute inhalation toxicity		
Difluoromethane	:	LC50: > 520000 ppmExposure time: 4 h Species: Rat
Pentafluoroethane	:	> 769000 ppmExposure time: 4 h Species: Rat
1,1,1,2-Tetrafluoroethane	:	LC50: > 500000 ppm, gas Exposure time: 4 h Species: Rat

2,3,3,3-Tetrafluoroprop-1-ene	:	LC50: > 400000 ppm, vapour Exposure time: 4 h Species: Rat Method: OECD Test Guideline 403
trans-1,3,3,3-Tetrafluoroprop-1-ene	:	100000 ppmSpecies: Mouse Note: Acute (4-Hour) Inhalation Toxicity Screening Study (mouse): No lethality at >100,000 ppm. LC50: > 207000 ppmExposure time: 4 h Species: Rat
Acute dermal toxicity	:	Note: No data available

Skin corrosion/irritation

2,3,3,3-Tetrafluoroprop-1-ene	:	Note: Not applicable study technically not feasible
trans-1,3,3,3-Tetrafluoroprop-1-ene	:	Species: Rabbit Result: No skin irritation Method: OECD Test Guideline 404

Serious eye damage/eye irritation

2,3,3,3-Tetrafluoroprop-1-ene	:	Note: Not applicable study technically not feasible
Respiratory or skin sensitisation Difluoromethane	:	Cardiac sensitization Species: dogs Note: No-observed-effect level >350 000 ppm
Respiratory or skin sensitisation Pentafluoroethane	:	Cardiac sensitization Species: dogs Note: No-observed-effect level 75 000 ppm Lowest observed effect level

100 000 ppm

Respiratory sensitisation 1,1,1,2-Tetrafluoroethane	:	Cardiac sensitization Species: dogs Note: No-observed-effect level 50 000 ppm Lowest observed effect level 75 000 ppm
Skin sensitisation 2,3,3,3-Tetrafluoroprop-1-ene	:	Note: Not applicable, as this product is a gas. study technically not feasible
Respiratory or skin sensitisation trans-1,3,3,3-Tetrafluoroprop-1-ene	:	Cardiac sensitization Species: dogs Note: Did not cause sensitisation on laboratory animals.

Repeated dose toxicity

Difluoromethane	:	Species: Rat Application Route: Inhalation Exposure time: (90 d) NOEL: 50000 ppm Subchronic toxicity
Pentafluoroethane	:	Species: Rat Application Route: Inhalation Exposure time: (4 Weeks) NOEL: 50000 ppm Subchronic toxicity
1,1,1,2-Tetrafluoroethane	:	Species: Rat NOEL: 40000 ppm
2,3,3,3-Tetrafluoroprop-1-ene	:	Species: Rat Application Route: Inhalation Exposure time: (2 Weeks)

No-observed-effect level: 50000 ppm
Method: OECD Test Guideline 412

Species: Rat
Application Route: Inhalation
Exposure time: (4 Weeks)
NOAEL (No observed adverse effect level): 50000 ppm
Method: OECD Test Guideline 412

Species: Rat
Application Route: Inhalation
Exposure time: (13 Weeks)
NOAEL (No observed adverse effect level): 50000 ppm
Method: OECD Test Guideline 413

Species: Rabbit, male
Application Route: Inhalation
Exposure time: (28 d)
No-observed-effect level: 500 ppm
Method: OECD Test Guideline 412
There are no observed toxicological effects, which result in classification as a specific target organ toxicant.

Species: Rabbit, female
Application Route: Inhalation
Exposure time: (28 d)
No-observed-effect level: 1000 ppm
Method: OECD Test Guideline 412
There are no observed toxicological effects, which result in classification as a specific target organ toxicant.

Species: Mini-pig
Application Route: Inhalation
Exposure time: (28 d)
NOAEL (No observed adverse effect level): 10000 ppm
highest exposure tested

trans-1,3,3,3- : Species: Rat

Tetrafluoroprop-1-ene		Application Route: Inhalation Exposure time: (13 Weeks) NOEL: 5000 ppm Species: Dog Application Route: Inhalation Exposure time: (9 Months) NOEL 14,700 ppm Species: Mouse Application Route: Inhalation Exposure time: (13 Weeks) NOEL 50,700 ppm
Germ cell mutagenicity		
Difluoromethane	:	Test Method: Ames test Result: negative
Pentafluoroethane	:	Test Method: Ames test Result: negative
1,1,1,2-Tetrafluoroethane	:	Note: In vitro tests did not show mutagenic effects
2,3,3,3-Tetrafluoroprop-1-ene	:	Test Method: Ames test Result: positive negative
trans-1,3,3,3-Tetrafluoroprop-1-ene	:	Test Method: Chromosome aberration test in vitro Cell type: Human lymphocytes Result: negative
	:	Cell type: Human lymphocytes Result: negative Method: Mutagenicity (in vitro mammalian cytogenetic test)
	:	Test Method: Chromosome aberration test in vitro Result: negative

: Cell type: Human lymphocytes
Result: negative

: Cell type: Chinese Hamster Ovary Cells
Result: negative

: Test Method: Chromosome aberration test in vitro
Cell type: Human lymphocytes
Result: negative
Method: OECD Test Guideline 473
Note: Dose 760,000 ppm

: Test Method: Ames test
Result: negative

Germ cell mutagenicity

Difluoromethane : Species: Mouse
Cell type: Bone marrow
Method: Mutagenicity (micronucleus test)
Result: negative

2,3,3,3-Tetrafluoroprop-1-ene : Test Method: Unscheduled DNA synthesis
Dose: up to 50,000 ppm (4 weeks)
Method: OECD Test Guideline 486
Result: negative

: Species: Rat
Cell type: Micronucleus
Dose: up to 50,000 ppm (4 weeks)
Method: OECD Test Guideline 474
Result: negative

trans-1,3,3,3-Tetrafluoroprop-1-ene : Test Method: Mutagenicity (in vivo mammalian bone-marrow cytogenetic test, chromosomal analysis)
Species: Mouse
Cell type: Micronucleus
Application Route: Inhalation
Result: negative

Carcinogenicity 2,3,3,3-Tetrafluoroprop-1-ene	:	Species: Rat Note: Not classified as a human carcinogen. Substance not expected to be a carcinogen based on available data.
trans-1,3,3,3-Tetrafluoroprop-1-ene	:	Species: Mouse Note: Did not show carcinogenic effects in animal experiments. Species: Rat Note: Did not show carcinogenic effects in animal experiments.
Reproductive toxicity 2,3,3,3-Tetrafluoroprop-1-ene	:	Species: Rat Application Route: Inhalation exposure Exposure time: Two-generation reproductive toxicity NOAEL,parent: 50,000 ppm NOAEL,F1: 50,000 ppm NOAEL,F2: 50,000 ppm
Teratogenicity Difluoromethane	:	Species: Rat Dose: NOEL - 50,000 ppm Note: Did not show teratogenic effects in animal experiments. Species: Rabbit Dose: NOEL - 50,000 ppm Note: Did not show teratogenic effects in animal experiments.
Pentafluoroethane	:	Species: Rabbit Application Route: Inhalation exposure NOAEL,Teratog: 50,000 ppm

NOAEL,Maternal: 50,000 ppm
Note: Did not show teratogenic effects in animal experiments.

Species: Rat
Application Route: Inhalation exposure
NOAEL,Teratog: 50,000 ppm
NOAEL,Maternal: 50,000 ppm
Note: Did not show teratogenic effects in animal experiments.

2,3,3,3-Tetrafluoroprop-1-ene	:	Species: Rat Dose: NOAEL (No observed adverse effect level) - 50,000 ppm Species: Rabbit Dose: NOAEL (No observed adverse effect level) - 4,000 ppm
trans-1,3,3,3-Tetrafluoroprop-1-ene	:	Species: Rabbit Method: Prenatal Developmental Inhalation Toxicity Study Note: Did not show teratogenic effects in animal experiments. Species: Rat Method: Prenatal Developmental Inhalation Toxicity Study Note: Did not show teratogenic effects in animal experiments.
Specific target organ toxicity - single exposure	:	Note :No data available
Specific target organ toxicity - repeated exposure	:	Note :No data available
Further information 1,1,1,2-Tetrafluoroethane	:	Note: Vapours are heavier than air and can cause suffocation by reducing oxygen available for breathing. Rapid evaporation of the liquid may cause frostbite. Avoid skin contact with leaking liquid (danger of frostbite).
trans-1,3,3,3-Tetrafluoroprop-1-ene	:	Note: Cardiac NOEL 119, 826 ppm.

12. ECOLOGICAL INFORMATION

Aquatic and ecotoxicity

Toxicity to fish

2,3,3,3-Tetrafluoroprop-1-ene	:	LC50: > 197 mg/l Exposure time: 96 h Species: Cyprinus carpio (Carp) Method: OECD Test Guideline 203 Note: No demonstrable toxic effect in saturated solution.
trans-1,3,3,3-Tetrafluoroprop-1-ene	:	NOEC: > 117 mg/l Exposure time: 96 h Species: Cyprinus carpio (Carp)

Toxicity to daphnia and other aquatic invertebrates

2,3,3,3-Tetrafluoroprop-1-ene	:	EC50: > 83 mg/l Exposure time: 48 h Species: Daphnia magna (Water flea) Method: OECD Test Guideline 202
trans-1,3,3,3-Tetrafluoroprop-1-ene	:	EC50: > 160 mg/l Exposure time: 48 h Species: Daphnia magna (Water flea)

Toxicity to algae

2,3,3,3-Tetrafluoroprop-1-ene	:	EC50: > 100 mg/l Species: Scenedesmus capricornutum (fresh water algae) Method: OECD Test Guideline 201
trans-1,3,3,3-Tetrafluoroprop-1-ene	:	Growth inhibition NOEC: > 170 mg/l Exposure time: 72 h Species: Algae

Persistence and degradability

Biodegradability
Difluoromethane : Note: Minimal

Pentafluoroethane : Result: Not readily biodegradable.
Value: 5 %
Method: OECD 301 D

2,3,3,3-Tetrafluoroprop-1-ene : Result: Not readily biodegradable.
Method: OECD Test Guideline 301F

trans-1,3,3,3-Tetrafluoroprop-1-ene : aerobic
Result: Not readily biodegradable.

Bioaccumulative potential

Bioaccumulation
2,3,3,3-Tetrafluoroprop-1-ene : Note: Due to the distribution coefficient n-octanol/water, accumulation in organisms is not expected.

Mobility in soil

Mobility : Note: No data available

Other adverse effects
Additional ecological information
1,1,1,2-Tetrafluoroethane : Accumulation in aquatic organisms is unlikely.

13. DISPOSAL CONSIDERATIONS

Disposal methods : In accordance with local and national regulations.

14. TRANSPORT INFORMATION

ADR

UN/ID No.	:	UN 3163
Description of the goods	:	LIQUEFIED GAS, N.O.S. (PENTAFLUOROETHANE, DIFLUOROMETHANE, 1,1,1,2-TETRAFLUOROETHANE)
Class	:	2
Classification Code	:	2A
Hazard Identification Number	:	20
Labels	:	2.2

IATA

UN/ID No.	:	UN 3163
Description of the goods	:	Liquefied gas, n.o.s. (Pentafluoroethane, Difluoromethane, 1,1,1,2-Tetrafluoroethane)
Class	:	2.2
Labels	:	2.2
Packing instruction (cargo aircraft)	:	200
Packing instruction (passenger aircraft)	:	200

IMDG

UN/ID No.	:	UN 3163
Description of the goods	:	LIQUEFIED GAS, N.O.S. (PENTAFLUOROETHANE, DIFLUOROMETHANE, 1,1,1,2-TETRAFLUOROETHANE)
Class	:	2.2
Labels	:	2.2
EmS Number 1	:	F-C
EmS Number 2	:	S-V
Marine pollutant	:	no

15. REGULATORY INFORMATION

National regulatory information

: No information available.

Other international regulations

Notification status

USA. List of Active Substances on the Toxic Substances Control Act (TSCA) Chemical Substances Inventory, as amended	:	All substances listed as active on the TSCA inventory
Australian Inventory of Industrial Chemicals	:	All components are listed on the inventory, regulatory obligations/restrictions apply
Canada. Domestic Substances List (DSL), as amended	:	All components of this product are on the Canadian DSL
Japan. Kashin-Hou Law List	:	On the inventory, or in compliance with the inventory
Korea. Existing Chemicals Inventory (KECI)	:	On the inventory, or in compliance with the inventory
Philippines. Inventory of Chemicals and Chemical Substances (PICCS)	:	Not in compliance with the inventory

China. Inventory of Existing Chemical Substances (IECSC)	:	On the inventory, or in compliance with the inventory
Taiwan Chemical Substance Inventory (TCSI)	:	On the inventory, or in compliance with the inventory

16. OTHER INFORMATION

		HMIS III	NFPA
Health hazard	:	1	2
Flammability	:	1	1
Physical Hazard	:	0	
Instability	:		0

Hazard rating and rating systems (e.g. HMIS® III, NFPA): This information is intended solely for the use of individuals trained in the particular system.

Further information

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text. Final determination of suitability of any material is the sole responsibility of the user. This information should not constitute a guarantee for any specific product properties.

Changes since the last version are highlighted in the margin. This version replaces all previous versions.
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Person who prepared the SDS Solstice Product Stewardship Group